Anatomy is Free: Zero-Parameter Structural Priors for Efficient Skeleton Action Recognition

Supplementary Material

Anonymous WACV Applications Track submission

Paper ID 2586

This supplement contains the material referenced from the main paper: the partitions KATP (Kinematic-Tree Anatomical Partition) derives for skeleton layouts other than NTU-25 (Section 1), the accuracy–compute frontier plotted explicitly (Section 2), and the block-level and hyperparameter ablations for which the main text has no room (Sections 3–4). Section 5 lists the three model configurations in full. No result here is new; every figure is as reported in the main paper.

1. KATP on Other Skeleton Layouts

KATP reads a bone list and nothing else, so it applies unchanged to layouts with different joint counts and different conventions. Table 1 gives the partitions it returns for the 20-joint Kinect v1 layout used by NW-UCLA and for the 17-keypoint COCO layout produced by 2-D pose estimators, alongside NTU-25 for reference.

Two points are worth noting. First, the COCO bone graph is *not* a tree: the shoulder–shoulder, hip–hip and facial links form cycles, so step 1 of the derivation reduces it to its breadth-first spanning tree from the root. The branch joints are then the two shoulders rather than a single spine hub, and the bilateral pairing rule matches limbs across mirrored anchors of equal depth. Second, COCO has no pelvis keypoint, so the hips fall inside the leg chains, as they do on NTU-25.

Region names follow from anchor depth and are cosmetic; the routing operators consume the groups, never the labels.

2. Accuracy–Compute Frontier

Figure 1 plots the GFLOPs and NTU-60 X-Sub columns of the main results table. All three models lie on the frontier, and SFZ-Nano occupies a compute budget an order of magnitude below the efficient baselines.

Table 1. Partitions derived by KATP from the bone list alone. The NTU-25 rows are identical to the hand-written grouping used throughout the main paper. We make no accuracy claim on the other two layouts: this demonstrates generality of the *construction* only.

Layout	Region	Joints
	left arm	4, 5, 6, 7, 21, 22
	right arm	8, 9, 10, 11, 23, 24
NTU-25	left leg	12, 13, 14, 15
	right leg	16, 17, 18, 19
	torso	0, 1, 2, 3, 20
	left arm	4, 5, 6, 7
	right arm	8, 9, 10, 11
NW-UCLA-20	left leg	12, 13, 14, 15
	right leg	16, 17, 18, 19
	torso	0, 1, 2, 3
	left arm	7, 9
	right arm	8, 10
COCO-17	left leg	11, 13, 15
	right leg	12, 14, 16
	torso	0, 1, 2, 3, 4, 5, 6

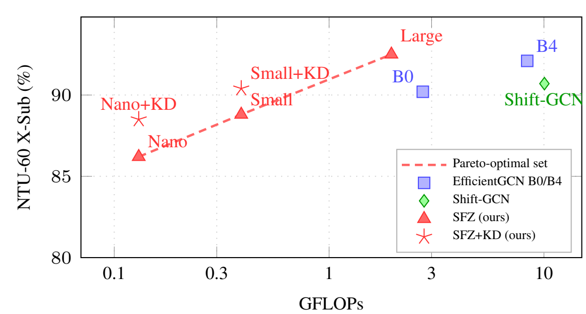

Figure 1. Accuracy against compute on NTU-60 X-Sub (log FLOPs axis).

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3. Block Component Ablation

Table 2 removes or replaces one component of the EfficientZero block at a time. The first row after the full model is the block-ordering result quoted in the main paper: moving the two routing operators to *after* the graph convolution costs 1.9 pp, confirming that their value lies in pre-structuring the representation that the convolution then refines.

Replacing the fixed routing with a learned $V \times V$ soft assignment (625 parameters) costs 0.6 pp, so the derived permutation is not merely competitive with a cheap learned alternative — it is better, at zero cost.

Table 2. Block component ablation (SFZ-Nano, NTU-60 X-Sub). Each row removes or replaces one component of the full block.

Configuration	Params	Acc. (%)	Δ
Full EfficientZero block	94 K	**86.2**	—
Routing placed after the GCN	94 K	84.3	-1.9
Learned $V \times V$ routing (625 p)	94 K	85.6	-0.6
− STC-Attention	87 K	85.1	-1.1
− Joint embedding	87 K	85.2	-1.0
DS-TCN $\rightarrow$ standard TCN	101 K	84.9	-1.3
− DropPath (p_d=0)	93 K	85.0	-1.2

4. Graph and Regularisation Hyperparameters

Graph partition count. Table 3 varies the number of adjacency subsets K in the spatial graph convolution. Moving from a single isotropic aggregation to the three-subset ST-GCN partition is worth 1.2 pp. The final row enables the learned residual adjacency $\mathbf{A}_\ell$ and TLA together, so its value is not comparable with the rows above it.

Table 3. Graph partition count (SFZ-Nano, NTU-60 X-Sub). TLA is disabled except in the last row.

K	$\mathbf{A}_\ell$	Params	Acc. (%)
1	—	87 K	83.4
2		90 K	84.2
3		92 K	84.6
3	✓	94 K	**86.2**

Stochastic depth. Table 4 sweeps the DropPath rate at all three scales. The two smaller models prefer a moderate rate; SFZ-Large needs a higher one to regularise 1.18 M parameters over 150 epochs. TLA is withheld from the SFZ-Nano sweep and PartAttention from the SFZ-Small sweep so that p_d is isolated, which is why the column maxima sit below the full-model figures reported in the main paper.

Stream fusion. Table 5 compares where the streams are combined in SFZ-Large. Mid-network fusion beats late fusion by 0.6 pp while using 40% fewer parameters: shared stages are more parameter-efficient than independent per-stream backbones.

Table 4. DropPath rate p_d (NTU-60 X-Sub).

p_d	SFZ-Nano	SFZ-Small	SFZ-Large
0.00	84.9	86.8	91.6
0.05	85.2	87.2	91.9
0.10	**85.9**	**87.6**	92.1
0.15	85.4	87.4	92.3
0.20	85.1	87.1	**92.5**
0.25	84.8	86.7	92.3

Table 5. Fusion strategy for SFZ-Large (NTU-60 X-Sub).

Fusion strategy	Params	X-Sub
Early (single backbone, 4 streams)	0.67M	89.8
Late (3 independent backbones)	1.93M	91.9
Mid-network (ours, B4-style)	**1.18M**	**92.5**

Cross-stream fusion. Table 6 ablates the gated exchange between the two backbones of SFZ-Small.

Table 6. Cross-stream fusion in SFZ-Small (NTU-60 X-Sub).

CSF variant	Params	X-Sub
Gated (ours)	**281 K**	**88.8**
Plain concatenation, no gate	281 K	88.4
No cross-stream fusion	271 K	88.0

5. Model Configurations

Table 7. Variant configurations. C_s stem channels, C stage widths, N blocks per stage, K TLA landmarks, r reduction ratio, p_d DropPath rate. †SFZ-Large uses B4-style mid-network fusion: per-stream 64→96→48 (depth 2+2), concatenation to 144 channels, then shared 144→128→272 (depth 3+3).

Model	C_s	C	N	K	r	p_d	Params
SFZ-Nano	24	[32,64,128]	[1,1,1]	12	8	0.10	94 K
SFZ-Small	24	[32,64,128]	[1,2,1]	8	8	0.10	281 K
SFZ-Large†	64	see †	—	14	8	0.20	1.18 M

Training. All models are trained from scratch with SGD, Nesterov momentum 0.9, and a cosine schedule with a 10-epoch linear warm-up. SFZ-Nano and SFZ-Small: learning rate 0.1, weight decay 4×10^{-4}, batch 64, 300 epochs, Mixup α=0.1. SFZ-Large: weight decay 10^{-4}, batch 16, 150 epochs, no Mixup, label smoothing 0.1. Weight decay is not applied to normalisation parameters, joint embeddings, $\mathbf{A}_\ell$, TLA anchor logits, or any gate scalar. Distillation, where used, has $\tau=4$ and $\lambda=0.5$ with a frozen SFZ-Large teacher.